User Manual FFT DataBridge

v. 3.0.0-release

as of 16.06.2026

Table of Contents

Disclaimer of liability

This document serves as a user guide for the use of the **FFT DataBridge**. It does not replace Siemens' official product and IIH documentation.

Scope / Versions: The content reflects the features available at the time of writing for the FFT DataBridge and the IIH Essentials OpenAPI (≥ 2.2.0). Interfaces, menus, and screenshots may vary depending on IEM, firmware, and app versions. Subject to change without notice

Responsibility: Installation, configuration, operation, and network security (in particular access rights, port exposure, certificates) are the responsibility of the operator. Ensure compliance with all regulatory, data protection, and IT security requirements (e.g., GDPR) and maintain appropriate backups.

Third-Party Dependencies: Examples include "Industrial Information Hub (IIH)" and "Snowflake," which are trademarks and/or products of their respective owners. References to external products are solely for integration purposes and do not constitute an endorsement.

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1. Preface

1.1. Introduction

FFT DataBridge solution empowers manufacturing companies to securely, scalably, and efficiently move production data into the cloud. Targeted at production managers, data engineers, and IT leaders. This solution addresses common challenges like data silos, limited transparency, and inefficient analytics.
With a standardized interface architecture, it connects shopfloor systems directly to the Snowflake or Databricks AI Data Cloud. Customers benefit from real-time analytics, AI-based forecasting, and reduced downtime. The solution delivers measurable value through faster decision-making, higher data quality, and optimized IT costs — a key enabler for data-driven Smart Manufacturing.

1.2. Target Audience

This guide is intended for technical professionals with prior experience in Industrial Information Hub (IIH) and Snowflake or Databricks Data Cloud environments. The intended audience includes:

- Data engineers
- Automation engineers
- Edge and Cloud integration specialists
- IT personnel responsible for industrial data pipelines and cloud analytics.

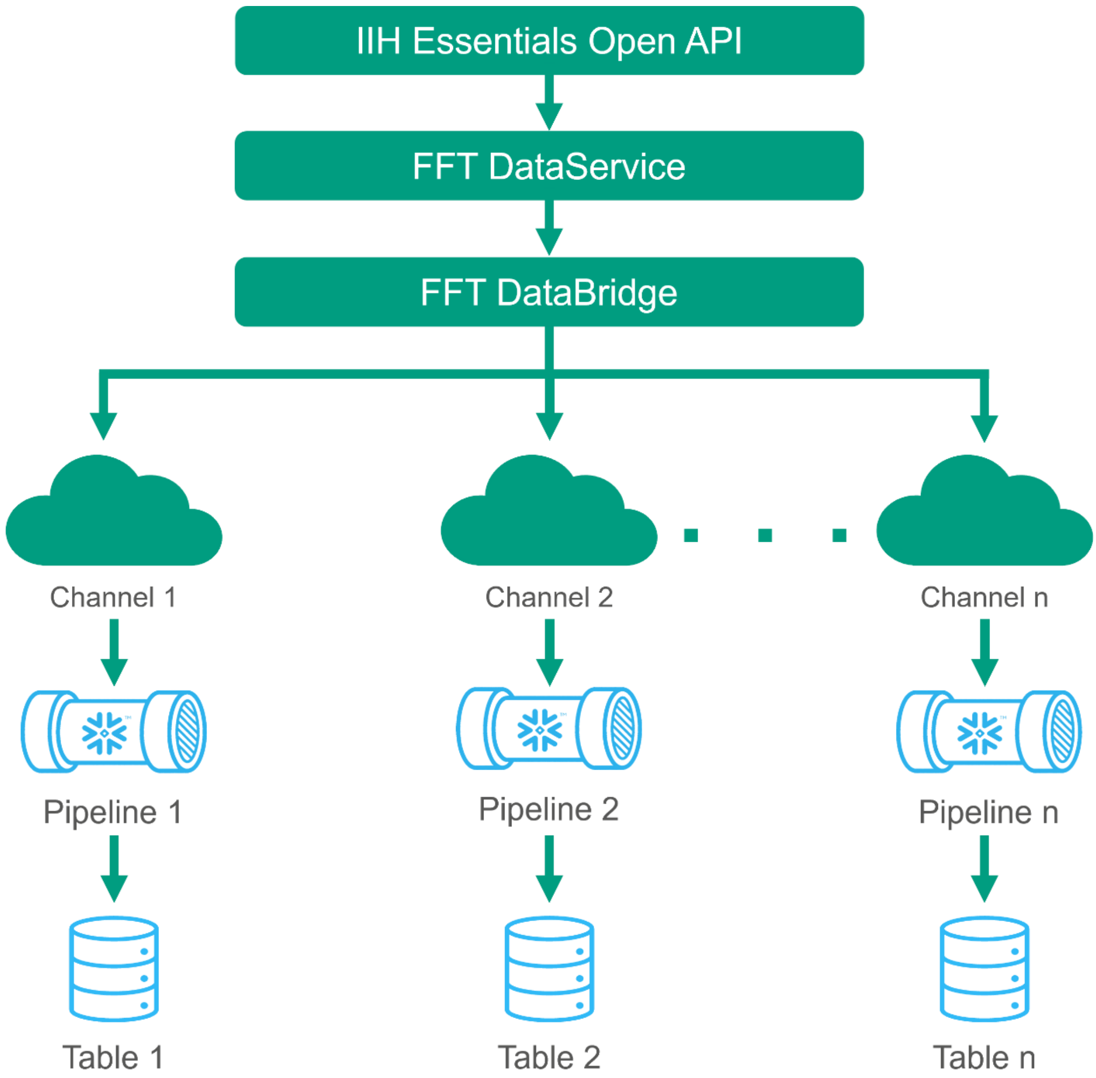
Familiarity with the following is expected:

- IIH components (e.g. Databus, Semantics, Common Configurator),
- Edge device deployment and configuration using Industrial Edge Management (IEM),
- Basic Snowflake or Databricks concepts such as warehouses, roles, stages, pipes, and authentication mechanisms.

This guide does **not** provide introductory explanations for Snowflake,Databricks or IIH. It assumes a working knowledge of these systems, focusing instead on the configuration and deployment of the **FFT DataBridge** in a production-grade environment.

2. Overview

The FFT DataBridge works in combination with the FFT-developed DataService. The Data Service is available in the Siemens Marketplace. The infrastructure works as follows. The DataService communicates with the IIH Essentials Open API and returns the requested data in the configured format back to the FFT Data Bridge. The DataBridge sends the data to Snowflake or Databricks and transfers the data into the preconfigured Tables. The IIH Essentials Open API must be at least **version 2.2.0** or higher**.**



3. Installing the App

3.1. Software Requirements

- IIH Essentials
- FFT DataService

3.2. Hardware Requirements

- Running Industrial Edge Management (IEM) instance is required. Tested with IEM ISO 1.15.7 and later; other versions may be compatible.
- An Edge Device that is compatible with Industrial Edge Management
    - IED Model e.g. Simatic IPC427 E, IPC227G,
    - Hard disk at least 2 GB
    - RAM: 500 MB RAM available

3.3. Install the App itself

[How to install an Industrial Edge App? - ID: 109824882 - Industry Support Siemens]

3.4. Target Platform Configuration

The FFT DataBridge can be configured to send data to either a Snowflake or a Databricks environment.

Only a single target platform can be configured per DataBridge instance. Concurrent operation with both platforms is not supported.

4. Configuring Snowflake

The configuration is split into two parts, the first part is the Snowflake configuration, in this part we prepare the Snowflake, Pipeline, Table and the required privileges. The second part is the configuration for the app itself on the edge device.

It is recommended to create an individual user only for the FFT DataBridge with a role that follows the least privilege principle. The user only needs these rights and configurations:

- Key Value Pair Authentication configured see Snowflake tutorial here for encrypted private key without MFA: (Key-pair authentication and key-pair rotation | Snowflake Documentation)
- USAGE ON DATABASE "*DATABASE_NAME*"
- USAGE ON SCHEMA "*SCHEMA_NAME*"
- INSERT, SELECT on "*TABLE_NAME*"
- OPERATE, MONITOR ON "*PIPE_NAME*"

Note: the Names are only examples

The template below outlines how the required Snowflake objects must be created. Parameters enclosed in angle brackets (e.g. <TABLENAME>) are placeholders and should be replaced with actual values.

Example template for the configuration:

```
USE DATABASE <DATABASE_NAME>;
USE SCHEMA <SCHEMA_NAME>;


--Here you can change the TABLE_NAME, the column definitions must not be -----
--changed
CREATE TABLE IF NOT EXISTS <TABLE_NAME>(
    ANCESTORNAME VARCHAR(16777216),
    VARIABLENAME VARCHAR(16777216),
    TIMESTAMP TIMESTAMP_NTZ(9),
    VALUE VARIANT
);


CREATE PIPE IF NOT EXISTS  <PIPE_NAME>
AS
COPY INTO <TABLE_NAME>
FROM TABLE(DATA_SOURCE(TYPE => 'STREAMING'))
MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE;


-- Required privileges: make sure your user has the following grants before --
--installing the FFT Data Bridge.
GRANT USAGE ON DATABASE <DATABASE_NAME> to ROLE <DATABRIDGE_ROLE_NAME>;
GRANT USAGE ON SCHEMA <DATABASE_NAME>.<SCHEMA_NAME> to ROLE
<DATABRIDGE_ROLE_NAME>;
GRANT INSERT, SELECT on Table <DATABASE_NAME>.<SCHEMA_NAME>.<TABLE_NAME> TO
ROLE <DATABRIDGE_ROLE_NAME>;
GRANT OPERATE, MONITOR ON PIPE <DATABASE_NAME>.<SCHEMA_NAME>.<PIPE_NAME> to
ROLE <DATABRIDGE_ROLE_NAME>;
```

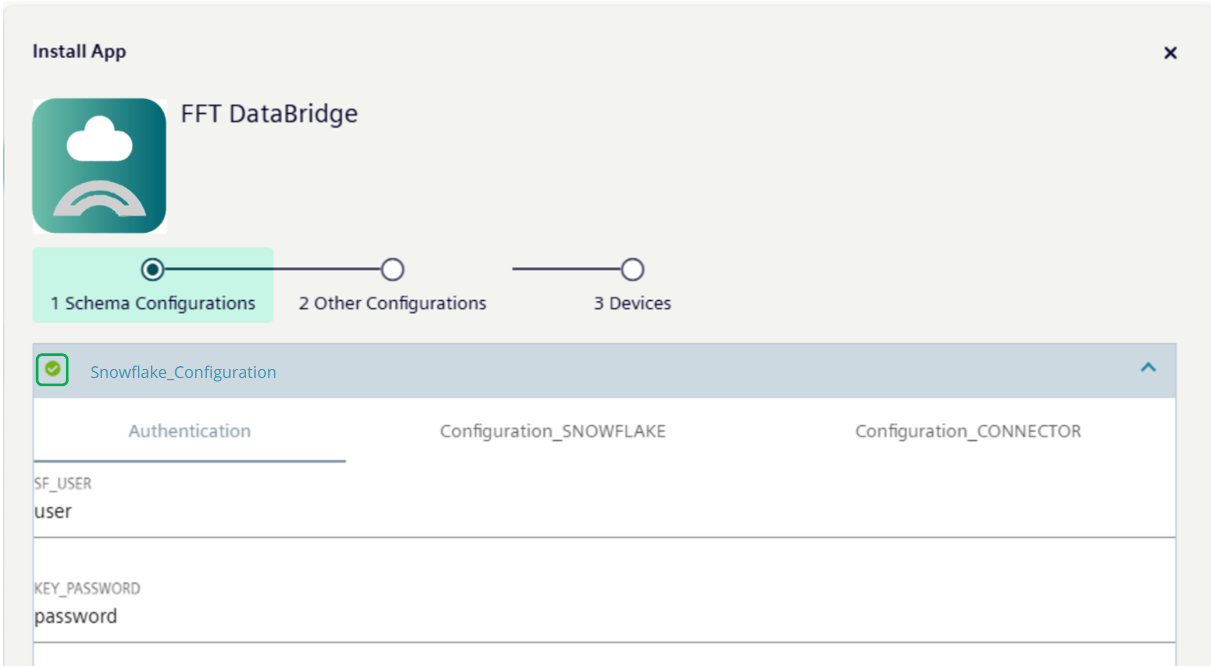
5. Configuring DataBridge for Snowflake

There are two different configuration modes. The first mode is intended for standard users who simply want to transfer all data available in the IIH API to the Snowflake Cloud with minimal setup.

The second mode is designed for expert users who require fine-grained control over which data is written to which table. This mode offers advanced filtering options but comes with increased configuration complexity.

5.1. Standard-Setup

The following explanations are for the standard user. It is very important to click the green checkmark to activate the Snowflake_Configuration.

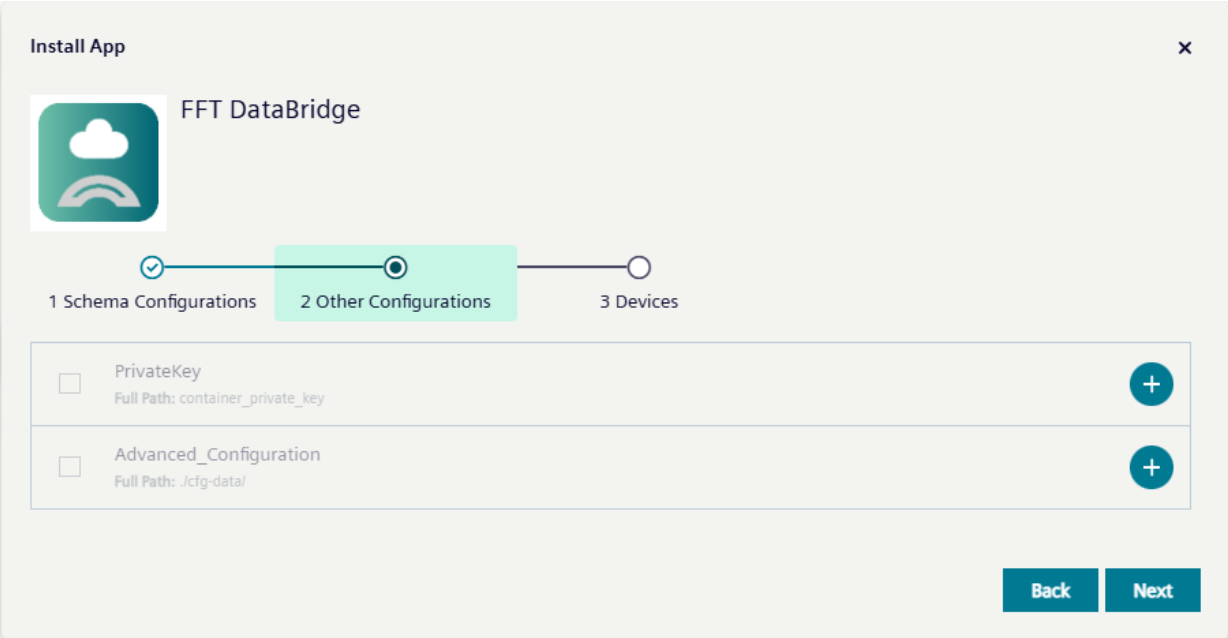


Authentication		
Attribute	**Default**	**Description**
SF_USER	“”	Username for Data Bridge User
KEY_PASSWORD	“”	Password for encrypted Private key

Configuration_SNOWFLAKE		
Attribute	**Default**	**Description**
ACCOUNT	<account-identifier>	Account to which connector should connect
Snowflake_Server_URL	<account-identifier> .snowflakecomputing.com>	Complete Account/Server URL from Snowflake
PIPE	<PIPENAME>	Created pipe for connector
SCHEMA	<SCHEMANAME>	Database schema
DATABASE	<DATABASENAME>	Database where data should be stored

Configuration_CONNECTOR		
Attribute	**Default**	**Description**
QUERY_INTERVAL	15	The interval in which data is fetched from FFT Data Service. Minimum is 1 second.
START_TIMESTAMP_OVERRIDE	“”	Starttime where the DataBridge starts the loading of the history data. When none is given all data is fetched. The format is the following: YYYY-MM-DDTHH:MM:SS.000000Z

The second step involves uploading the private key file that is configured for the Snowflake user. It is important to convert the file to ***.txt** format before uploading, as the file upload feature does not allow ***.p8** or ***.pem** files. Files must not contain multiple extensions. For example, rename **rsa_key.p8** to **rsa_key.txt**. In the latest IEM version, it is possible to upload the *.p8 files directly. This feature remains in place to maintain downward compatibility.



It is essential that the FFT DataService is installed before deploying the FFT DataBridge. Once the DataService is running and the device_list file can be downloaded, the FFT DataBridge may be installed. After installation, all variables available in the DataService as listed in the device list will be transferred to the target table.

If you want to switch the configuration mode, you must uninstall the application and reinstall it following the guidelines for the advanced setup. Vice versa.
Note as soon as the checkmark is set and the app is updated or uploaded to the device, the IEM automatically generates a configuration, which overrides an existing ***.toml** configuration.

5.2. Advanced-Setup

This section describes the advanced rule-based routing mechanism used to map incoming datapoints to specific Snowflake tables.
It is intended for expert users who require fine-grained control over how data is filtered and stored. The configuration does not use the UI schema configuration used in the standard setup, to maintain full flexibility the following *.**toml** file configures every step.

The Configuration is split into four parts. The Logging configuration, snowflake base configuration, IIH configuration and pipeline filter configuration.

5.2.1. Logging Configuration

[logging]		
Attribute	**Default**	**Description**
console_log	true	Enables/disables console logging.
level	INFO	Defines log level (INFO, WARNING, ERROR).
file_backup_count	3	Number of rotated log files to keep.

5.2.2. Snowflake Base Configuration

[snowflake]		
Attribute	**Default**	**Description**
SF_USER	“”	Snowflake User
ACCOUNT	“”	Snowflake Account
HOST	“”	Snowflake URL
KEY_PASSWORD	“”	Password for encrypted private key

5.2.3. IIH Configuration

[iih]		
Attribute	**Default**	**Description**
Data_Service_Container_Name	“”	Default name and port is set automatically. If you have changed the port in the DataService configuration you have to adapt this value.
QUERY_INTERVALL	15	The interval in which data is fetched from FFT Data Service. Minimum is 1 second
START_TIMESTAMP_OVERRIDE	“”	Starttime where the DataBridge starts the loading of the history data. When none is given all data is fetched. The format is the following: YYYY-MM-DDTHH:MM:SS.000000Z

5.2.4. Pipeline Filter Configuration

[pipelines.<name>]		
Attribute	**Default**	**Description**
DATABASE	“”	Snowflake Database
SCHEMA	“”	Snowflake Schema
PIPE	“”	Snowflake Snowpipe
MATCH	“”	Match Filter

The filter mechanism is explained in chapter 8.

5.2.5. Upload File to IEM

To activate the configured setup, you must skip the first part of the wizard and must not select the checkbox for the default configuration.

Under 2. Other Configuration, drag and drop or double-click the corresponding field. The ***.toml** file is uploaded to the Configuration field, and the ***.txt** private key file is uploaded to the Private Key field.

5.2.6. Full Example config.toml

The template below outlines how the required Databricks objects must be created. Parameters enclosed in angle brackets (e.g. <TABLENAME>) are placeholders and should be replaced with actual values.

```
[logging]
console_log = true
level = "INFO"
file_backup_count = 3

[snowflake]
SF_USER = "<DataBridge_User>"
ACCOUNT = "<Snowflake_Account>"
HOST = "<Snowflake_Account>.snowflakecomputing.com"
KEY_PASSWORD = "<PrivateKey_Password>"

[iih]
Data_Service_Container_Name = ""
QUERY_INTERVALL = "15"
START_TIMESTAMP_OVERRIDE = ""


[pipelines]
[pipelines.<FILTERNAME_1>]
DATABASE = "<DATABASE_NAME_1>"
SCHEMA = "<SCHEMA_NAME>"
PIPE = "<PIPE_NAME_1>"
MATCH = "<FILTER_1>"


[pipelines.<FILTERNAME_2>]
DATABASE = "<DATABASE_NAME_2>"
SCHEMA = "<SCHEMA_NAME>"
PIPE = "<PIPE_NAME_2>"
MATCH = "<FILTER_2>"

[pipelines.<FILTERNAME_3>]
DATABASE = "<DATABASE_NAME_3>"
SCHEMA = "<SCHEMA_NAME>"
PIPE = "<PIPE_NAME_3>"
MATCH = "<FILTER_3>"
```

6. Configuring Databricks

The configuration is split into two parts, the first part is the Databricks configuration, in this part we prepare the Databricks, Pipeline, Table and the required privileges. The second part is the configuration for the app itself on the edge device.

It is recommended to create an individual user only for the FFT DataBridge with a role that follows the least privilege principle. The user only needs these rights and configurations make sure you are using managed delta tables, default storage is not supported:

• Service principal with configured OAuth Secret in the Databricks UI.
• USE CATALOG ON CATALOG “CATALOG_NAME”
• USE SCHEMA ON SCHEMA“SCHEMA_NAME”
• MODIFY, SELECT ON TABLE “TABLE_NAME”
Note: the Names are only examples

The template below outlines how the required Databricks objects must be created. Parameters enclosed in angle brackets (e.g. <TABLE_NAME>) are placeholders and should be replaced with actual values.

Example template for the configuration:

```
--Here you can change the TABLE_NAME, the column definitions must not be -----
--changed
CREATE or replace TABLE <CATALOG_NAME>.`<SCHEMA_NAME>`.<TABLE_NAME>(
  timestamp STRING,
  value VARIANT,
  ancestorName STRING,
  variableName STRING);

-- Required privileges: make sure your user has the following grants before --
--installing the FFT Data Bridge.
GRANT USE CATALOG ON CATALOG <CATALOG_NAME> TO `<SERVICE_PRINCIPAL_UUID>`;
GRANT USE SCHEMA ON SCHEMA <CATALOG_NAME>.`<SCHEMA_NAME>` TO
`<SERVICE_PRINCIPAL_UUID>`;
GRANT MODIFY, SELECT ON TABLE <CATALOG_NAME>.`<SCHEMA_NAME>`.<TABLE_NAME> TO
`<SERVICE_PRINCIPAL_UUID>`;
```

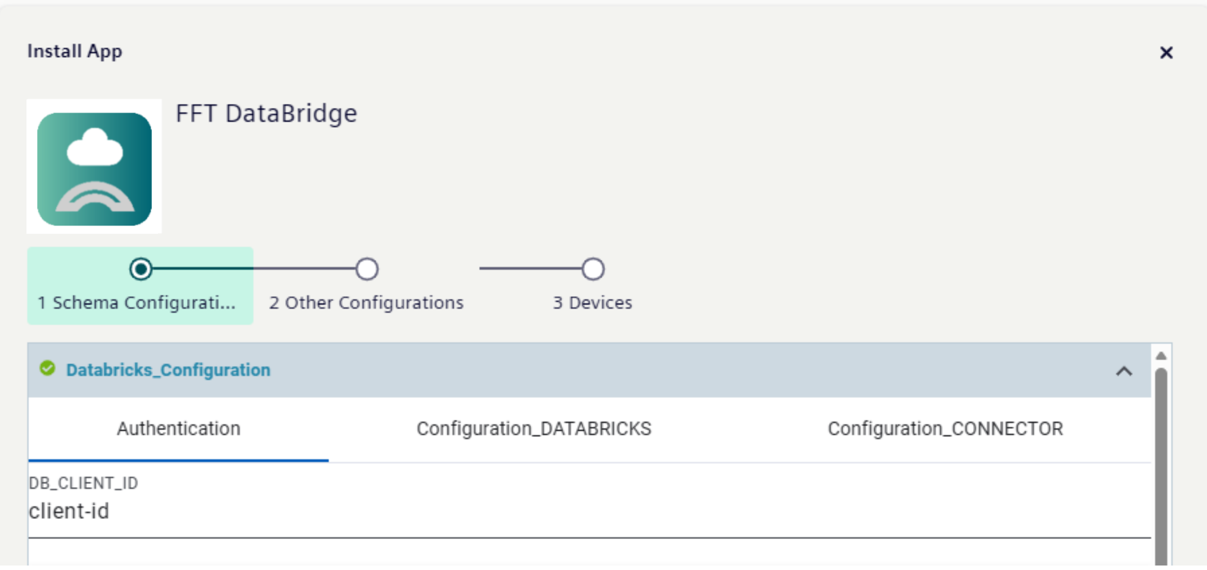
7. Configuring DataBridge for Databricks

There are two different configuration modes. The first mode is intended for standard users who simply want to transfer all data available in the IIH API to the Databricks Cloud with minimal setup.

The second mode is designed for expert users who require fine-grained control over which data is written to which table. This mode offers advanced filtering options but comes with increased configuration complexity

7.1. Standard-Setup

The following explanations are for the standard users. It is very important to click the green checkmark to activate the Databricks_Configuration.

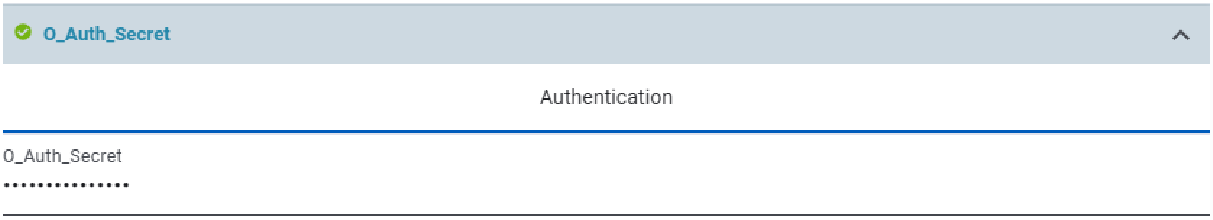


Authentication		
Attribute	**Default**	**Description**
DB_CLIENT_ID	“”	UUID for service principal

Configuration_DATABRICKS		
Attribute	**Default**	**Description**
WORKSPACE_URL	< https://<hostname-prefix>.cloud.databricks.com>	Databricks URL
SERVER_ENDPOINT	<worspaceID>.zerobus.<cloud-region>.cloud.databricks.com>	Complete server endpoint URL from Databricks
CATALOG	<catalog-name>	Created catalog
SCHEMA	<schema-name>	Database schema
TABLE	<table-name>	Database where data should be stored

Configuration_CONNECTOR		
Attribute	**Default**	**Description**
QUERY_INTERVAL	15	The interval in which data is fetched from FFT Data Service. Minimum is 1 second.
START_TIMESTAMP_OVERRIDE	“”	Starttime where the DataBridge starts the loading of the history data. When none is given all data is fetched. The format is the following: YYYY-MM-DDTHH:MM:SS.000000Z

The second step is to add the client secret to the DataBridge. To activate the secret, click the green checkmark next to O_Auth_Secret. Ensure that no whitespace characters are added when copying the secret.



It is essential that the FFT DataService is installed before deploying the FFT DataBridge. Once the DataService is running and the device_list file can be downloaded, the FFT DataBridge may be installed. After installation, all variables available in the DataService as listed in the device list will be transferred to the target table.

If you want to switch the configuration mode, you must uninstall the application and reinstall it following the guidelines for the advanced setup. Vice versa.

Note as soon as the checkmark is set and the app is updated or uploaded to the device, the IEM automatically generates a configuration, which overrides an existing *.toml configuration.

7.2. Advanced-Setup

This section describes the advanced rule-based routing mechanism used to map incoming datapoints to specific Databricks tables.
It is intended for expert users who require fine-grained control over how data is filtered and stored. The configuration does not use the UI schema configuration used in the standard setup, to maintain full flexibility the following *.toml file configures every step.

The Configuration is split into four parts. The Logging configuration, Databricks base configuration, IIH configuration and pipeline filter configuration.

7.2.1. Logging Configuration

[logging]		
Attribute	**Default**	**Description**
console_log	true	Enables/disables console logging.
level	INFO	Defines log level (INFO, WARNING, ERROR).
file_backup_count	3	Number of rotated log files to keep.

7.2.2. Databricks Base Configuration

[databricks]		
Attribute	**Default**	**Description**
WORKSPACE_URL		Databricks URL
SERVER_ENDPOINT		Complete server endpoint URL from Databricks
CATALOG		Created catalog
SCHEMA		Database schema

7.2.3. IIH Configuration

[iih]		
Attribute	**Default**	**Description**
Data_Service_Container_Name	“”	Default name and port is set automatically. If you have changed the port in the DataService configuration you have to adapt this value.
QUERY_INTERVALL	15	The interval in which data is fetched from FFT Data Service. Minimum is 1 second
START_TIMESTAMP_OVERRIDE	“”	Starttime where the DataBridge starts the loading of the history data. When none is given all data is fetched. The format is the following: YYYY-MM-DDTHH:MM:SS.000000Z

7.2.4. Pipeline Filter Configuration

[pipelines.<name>]		
Attribute	**Default**	**Description**
CATALOG	“”	Databricks Catalog
SCHEMA	“”	Databricks Schema
TABLE	“”	Databricks Table
MATCH	“”	Match Filter

7.2.5. Upload Configuration file to IEM

To activate the configured setup, you must skip the first part of the wizard and must not select the checkbox for the Databricks_Configuration.

Under 2. Other Configuration, drag and drop or double-click the corresponding field. The ***.toml** file is uploaded to the Configuration field. The client-secret is upload as shown in the standard-setup.

7.2.6. Full Example config.toml

```
[logging]
console_log = true
level = "INFO"
file_backup_count = 3

[databricks]
DB_CLIENT_ID = "<UUID for service principal>"
WORKSPACE_URL = "<https://<hostname-prefix>.cloud.databricks.com>"
SERVER_ENDPOINT = "<worspaceID>.zerobus.<cloud-region>.cloud.databricks.com>"

[iih]
Data_Service_Container_Name = ""
QUERY_INTERVALL = "15"
START_TIMESTAMP_OVERRIDE = ""

[pipelines]
[pipelines.<FILTERNAME_1>]
CATALOG = "<CATALOG_NAME_1>"
SCHEMA = "<SCHEMA_NAME>"
TABLE = "<TABLE_NAME_1>"
MATCH = "<FILTER_1>"

[pipelines.<FILTERNAME_2>]
CATALOG = "<CATALOG_NAME_2>"
SCHEMA = "<SCHEMA_NAME>"
TABLE = "<TABLE_NAME_2>"
MATCH = "<FILTER_2>"

[pipelines.<FILTERNAME_3>]
CATALOG = "<CATALOG_NAME_3>"
SCHEMA = "<SCHEMA_NAME>"
TABLE = "<TABLE_NAME_3>"
MATCH = "<FILTER_3>"
```

8. Mapping Definition

The Mapping rules are based on the generated device_list. The Device list contains every variable accessible in the IIH. This File can be downloaded shortly after deploying the FFT DataService on the Edge Device. The FFT Data Bridge automatically fetches the device list. The file contains two variables to identify the Variable in the IIH.

- assetPath → full datapath in the defined Data Structure in IIH
- variable → Variable Name

To be able to filter every possible variable the filter merges these together. So, the filter always becomes
rulePath = assetPath.variableName

8.1. Mapping Rules

1. Exact Match

The rule must match the full asset path exactly.
Example: MATCH = "Factory.Line1"

- only matches this specific path
- highest priority

2. Prefix Match

The rule ends with # and matches all asset paths that start with the given prefix.
Example: MATCH = "Factory.Line1.#"

- Matches all paths that start with Factory.Line1., such as:
    - Factory.Line1.Robot1.speed
    - Factory.Line1.Robot9.Axis4.current
    - Factory.Line1.MotorA.temperature
- longer prefix wins over shorter (more specific match)

3. Wildcard Match

Wildcard	Meaning
*	exactly one path segment
#	multiple arbitrary segments
.#.	multiple nested segments

Example: MATCH =MATCH = "Factory.*.Robot*.temp*"

- Matches following patterns:
  - Factory.Area1.Robot3.temp
  - Factory.XYZ.Robot9.temperature_raw

Example: MATCH="Factory.#.Temperature"

- Matches following patterns
  - Factory.LineA.Sensor.Temperature
  - Factory.Temperature

5. Mixed Match Rules

Example: MATCH = "*.Robot.#”

- Matches following patterns:
  - Line1.Robot.Temperature”
  - Line2.Robot.Axis1.Torque”
  - Line2.Robot.Axis2.Torque”
- Different Match Rule types, can be combined to get desired variables into one table

5. Multiple Match Rules

Example: MATCH = ["Combo.Path1.#", "Combo.Path2.Sub#"]

- Matches following patterns:
  - Combo.Path1.Test.Value
  - Combo.Path2.SubSection.Deeper
- Multiple Match Rules can be applied to one Pipeline

6. Catch All Match

This rule is active, if the MATCH part for the pipeline definition is **MATCH= “”**. If not rule applies every datapoint is written in this Pipeline. That’s the same pipeline definition that’s used in the standard setup configuration. But **only one** Catch All Pipeline is **allowed** per DataBridge deployment.

8.1.1. Map Priority

The router processes MAP rules in strict order:

1. Exact Match
2. Prefix Match (longest prefix wins)
3. Wildcard Match
4. Catch-All Match

8.1.2. Troubleshooting

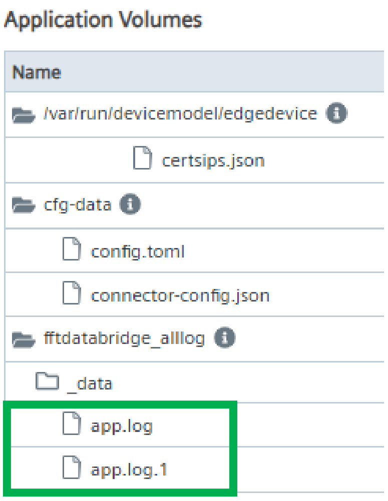
To simplify debugging, the filtered results generated by the mapping rules can be verified by downloading the device_list.json file from the DataBridge. This file contains all asset paths and variables that match a given filter.

To compare these with the total number of available asset paths and variables, the device list can be downloaded from the DataService and compared using a diff tool. Note that only variables that appear in the device list can be retrieved.

No datapoint can be stored in multiple tables, the first declared table that contains a matching filter gets the datapoint.

9. Logging

The Logs are divided in two files. These files are available in the Docker volume view in the edge device.



The app.log file contains application logs generated during the runtime of the application and can be accessed in the app view inside the Edge Device.

These files are stored using a ring-buffer mechanism, which can be adjusted in the advanced setup settings.

Logs can also be downloaded via the IIH Web UI. They are separated into two files:

- fft_databridge.log contains general startup logs and external logs.
- fft_databridge-error.log contains error logs

10. Limitations

The application uses the at-least-once principle, so duplicates can occur with unstable internet connections or unstable edge devices. Deduplication is performed inside the cloud environment.

When using low-resource devices, the variable query interval must not be set to a multiple of five seconds, such as 5, 10, 15, or 20 seconds. Other Siemens applications may use the same intervals, which can cause simultaneous queries and significantly increase the load on system resources.

11. Snowflake

11.1.1. Cost Monitoring Snowflake

The end customer is responsible for monitoring all costs resulting from the use of the FFT DataBridge. This includes any expenses related to data transfer and storage, such as pipeline usage and storage consumption. Data transfer costs cannot be monitored using resource monitors. Instead, use budgets to detect anomalies and monitor consumption. Monitor credit usage with budgets | Snowflake Documentation

11.1.2. Compatibility

The used Snowflake service is available in all Amazon Web Services (AWS), Microsoft Azure regions except for government-specific regions and regions in China.

The maximum number of PIPE objects configured for Snowpipe Streaming is by default limited to 1,000 per account and 10 per table.

11.2. Databricks

11.2.1. Cost Monitoring Databricks

The end customer is responsible for monitoring all costs associated with the use of the FFT DataBridge. This includes expenses related to data ingestion, data transfer, pipeline usage, and storage consumption.

- Zerobus Ingest System Tables

- billing system table

To specifically monitor Zerobus Ingest usage, apply the following filters:

billing_origin_product = 'LAKEFLOW_CONNECT'

product_features.lakeflow_connect.zerobus_request_type = 'GRPC'

11.2.2. Compatibility

- Data can only be written to managed Delta tables.
- Writing to storage resources protected by private endpoints is not supported.
- Table names are restricted to ASCII letters, numeric characters, and underscores (_).
- The Databricks workspace and the destination table must reside in the same supported Databricks region.

11.2.3. Potential Duplicate Records

The DataBridge only marks transmitted records as successfully processed after an acknowledgment (ACK) has been received from the target system.

If the application is terminated unexpectedly, the device is restarted, or a system failure occurs before the ACK is received and processed, the affected records remain pending. After recovery, these records are transmitted again to ensure that no data is lost.

As a result, duplicate records may occur in the target system. This behavior is intentional and follows an at-least-once delivery approach, prioritizing data integrity over the risk of data loss.

Potential duplicates can be identified and removed during downstream processing in Databricks.

12. Change history

Revision	Date	Name	Comment
1.0.0	22.09.2025	T.Dobritz	creation
2.0.0	17.12.2025	T.Dobritz	Refactored for the updated architecture.
3.0.0	16.06.2026	T.Dobritz	Added Databricks integration